

Instructions: All instructions remain the same from our previous assessments.

Derivatives

$$\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$$

$$\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$

$$\frac{d}{dx}(\sinh^{-1} x) = \frac{1}{\sqrt{1+x^2}}$$

$$\frac{d}{dx}(\cosh^{-1} x) = \frac{1}{\sqrt{x^2-1}}$$

$$\frac{d}{dx}(\tanh^{-1} x) = \frac{1}{1-x^2}$$

Moments and Centroid

$$M_x = \frac{\rho}{2} \int_a^b (f(x)^2 - g(x)^2) dx$$

$$M_y = \rho \int_a^b x(f(x) - g(x)) dx$$

$$m = \rho \int_a^b (f(x) - g(x)) dx$$

$$(\bar{x}, \bar{y}) = \left(\frac{M_y}{m}, \frac{M_x}{m} \right)$$

Integrals

$$\int u dv = uv - \int v du$$

$$\int \tan x dx = \ln |\sec x| + C$$

$$\int \sec x dx = \ln |\sec x + \tan x| + C$$

$$\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \left(\frac{x}{a} \right) + C$$

Polar Coordinates

$$\frac{dy}{dx} = \frac{\frac{dr}{d\theta} \sin \theta + r \cos \theta}{\frac{dr}{d\theta} \cos \theta - r \sin \theta}$$

$$A = \int_a^b \frac{1}{2} (f(\theta)^2 - g(\theta)^2) d\theta$$

$$L = \int_a^b \sqrt{r^2 + \left(\frac{dr}{d\theta} \right)^2} d\theta$$

Surface Area

$$A = \int_a^b 2\pi f(x) \sqrt{1 + f'(x)^2} dx$$

$$A = \int_a^b 2\pi x \sqrt{1 + f'(x)^2} dx$$

Area Between Curves

$$A = \int_a^b (f(x) - g(x)) dx$$

Work

$$W = \int_a^b F(x) dx$$

Hydrostatic Force

$$F = \rho g \int_a^b w(y) d(y) dy$$

Arc Length

$$L = \int_a^b \sqrt{1 + f'(x)^2} dx$$

Approximate Integration

$$|E_T| \leq \frac{K(b-a)^3}{12n^2}$$

$$|E_M| \leq \frac{K(b-a)^3}{24n^2}$$

$$|E_S| \leq \frac{L(b-a)^5}{180n^4}$$

Volumes

$$\text{Cross sections: } V = \int_a^b A(x) dx$$

$$\text{Cylindrical shells: } V = \int_a^b 2\pi x f(x) dx$$

Hyperbolic Functions

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

$$\tanh x = \frac{\sinh x}{\cosh x}$$

$$\sinh^{-1} x = \ln(x + \sqrt{x^2 + 1})$$

$$\cosh^{-1} x = \ln(x + \sqrt{x^2 - 1})$$

$$\tanh^{-1} x = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right)$$

Trig & Hyperbolic Identities

$$\cos^2 x + \sin^2 x = 1$$

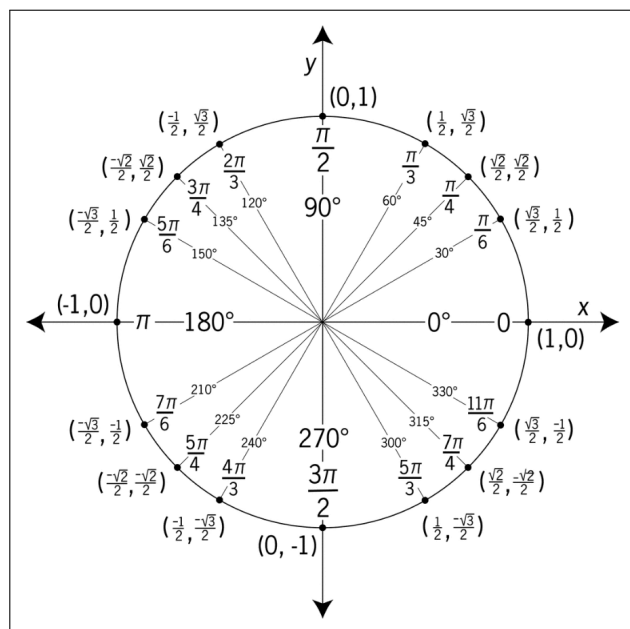
$$\tan^2 x + 1 = \sec^2 x$$

$$\sin^2 x = \frac{1}{2}(1 - \cos 2x)$$

$$\cos^2 x = \frac{1}{2}(1 + \cos 2x)$$

$$\cosh^2 x - \sinh^2 x = 1$$

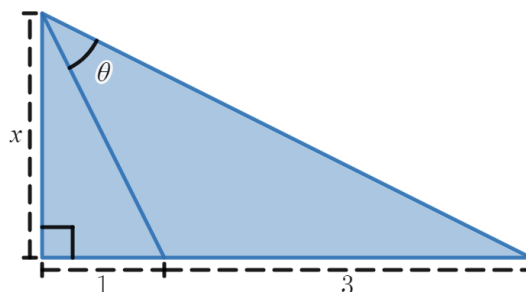
$$1 - \tanh^2 x = \operatorname{sech}^2 x$$



A1 - Inverse trigonometric and hyperbolic functions (inspired by Brightspace problem 4 and part 2 of the class notes)

(a) Consider the right triangle shown on the right.

- (i) Express θ as a function of x . *Hint:* θ is the difference of two angles whose tangents can be expressed using information from the diagram.



(b) Differentiate $f(x) = \arccos \ln(x)$.

(c) Recall that $\sinh(x) = \frac{e^x - e^{-x}}{2}$.

- (i) Show that $\sinh^{-1}(x) = \ln(x + \sqrt{x^2 + 1})$ by solving for y in the equation $x = \frac{e^y - e^{-y}}{2}$.

(ii) Use implicit differentiation with the equation $\sinh(y) = x$ to show $\frac{d}{dx} (\sinh^{-1}(x)) = \frac{1}{\sqrt{1 + x^2}}$.

A2 - L'Hospital's rule (inspired by Brightspace problems 2 and 10)

Evaluate each of the following limits.

(a) $\lim_{x \rightarrow 0} (1 + x^2 + x \sin(x))^{\frac{1}{x^2}}$

(b) $\lim_{x \rightarrow 0} x^{-4} \cdot \left(\cos(x) - 1 + \frac{x^2}{2} \right)$

A3 - Integration techniques I (inspired by part 1 of the class notes and past final worksheet problem 3)

Evaluate each of the following integrals. You do not need to simplify your answers.

(a) $\int_0^{0.5} \cos^{-1}(x) dx$

(b) $\int \frac{dx}{(4x^2 - 9)^{\frac{3}{2}}}$

A4 - Integration techniques II (inspired by part 1 of the class notes and past final worksheet problem 1)

For each of the following integrals, evaluate it or show that it is divergent.

(a) $\int \frac{x^5 + x^4 - 1}{x^4 + x^2} dx$

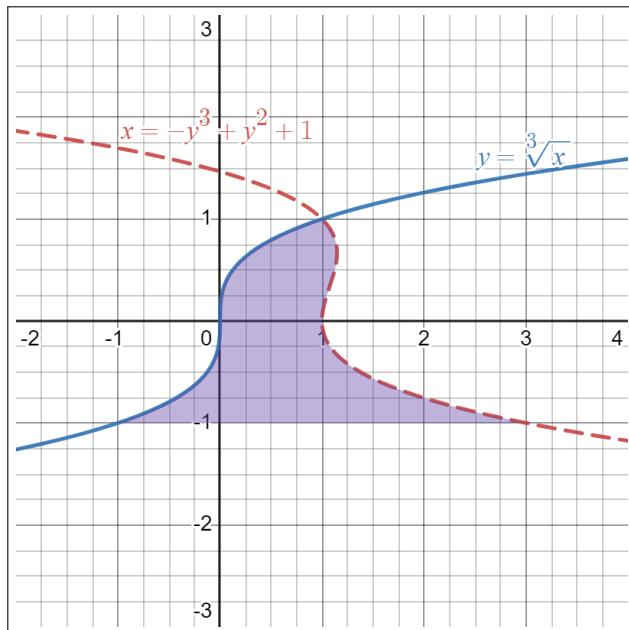
(b) $\int_{-\infty}^{\infty} \frac{e^x}{e^{2x} + 3} dx$

Consider the integral $I = \int_0^\pi \cos(2x)dx$, which we know is equal to 0 by straightforward antidifferentiation.

- (a) Use the midpoint rule with $n = 6$ to approximate I . You don't need to simplify your answer.
- (b) Find an upper bound on the error of your approximation from part (a) using techniques covered in class. You don't need to simplify your answer.
- (c) Use Simpson's rule with $n = 6$ to approximate I . You don't need to simplify your answer.
- (d) How large should n be to guarantee that your approximation from part (c) is accurate to within 0.00001. You don't need to simplify your answer.

B1 - Areas between curves (inspired by Brightspace problem 10 and the class notes)

- (a) Setup, but do not simplify or evaluate, an integral representing the area of the shaded region to the right. The dotted curve is $x = -y^3 + y^2 + 1$ and the solid curve is $y = \sqrt[3]{x}$.



- (b) Consider the area of the region bounded by $y = 3x^2$, $y = \frac{3}{x^2}$, $y = 0$, and $x = 3$. Find the number a such that the line $x = a$ bisects this area.

B2 - Volumes (inspired by Brightspace problem 5 and part 2 of the class notes)

(a) Consider the region enclosed by the parabola $y = x^2$ and the line $y = 4$.

(i) Setup, but do not evaluate, an integral representing the volume of the solid whose base is this region and whose cross sections perpendicular to the x -axis are squares.

(ii) Setup, but do not evaluate, an integral representing the volume of the solid whose base is this region and whose cross sections perpendicular to the y -axis are rectangles with height two times their base.

(b) Setup, but do not evaluate, an integral representing the volume of the solid obtained by rotating the region bound by $x = \sqrt{9 - y^2}$ and the y -axis about each of the following lines:

(i) $y = 3$

(ii) $x = -1$

B3 - Arclength and surface area (inspired by past final worksheet problem 2 and Brightspace problem 4)

(a) Let $f(x) = \left(4 - x^{\frac{2}{3}}\right)^{\frac{3}{2}}$.

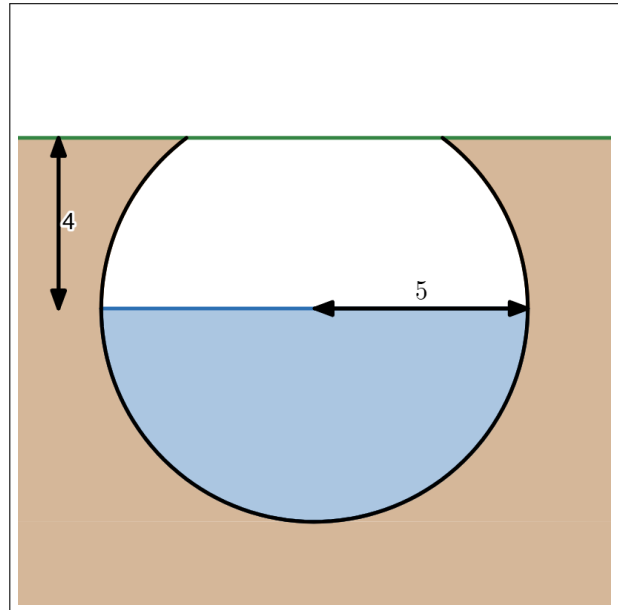
(i) Find the arclength function for $f(x)$ starting from $x = 0$.

(ii) Use the arclength function to find a number b such that the length of $f(x)$ from $x = 0$ to $x = b$ is 6.

(b) Find the surface area of the shape obtained by rotating the curve $y = \sqrt{1 + 2e^x}$, $0 \leq x \leq 1$, about the x -axis.

B4 - Applications of integration to physics and engineering (inspired by part 1 of the class notes and Brightspace problem 3)

- (a) A tank is the shape of a sphere of radius 5 meters with the top 1 meter cut off. This tank is buried underground so that the top of the tank is right at the surface, and the center of the sphere is 4 meters below the surface (shown to the right). If the tank is half full of a liquid of density ρ , setup an integral representing the work required to pump water to the surface until the depth of the water in the tank is 2 meters. You do not need to evaluate the integral. Use g for gravity.

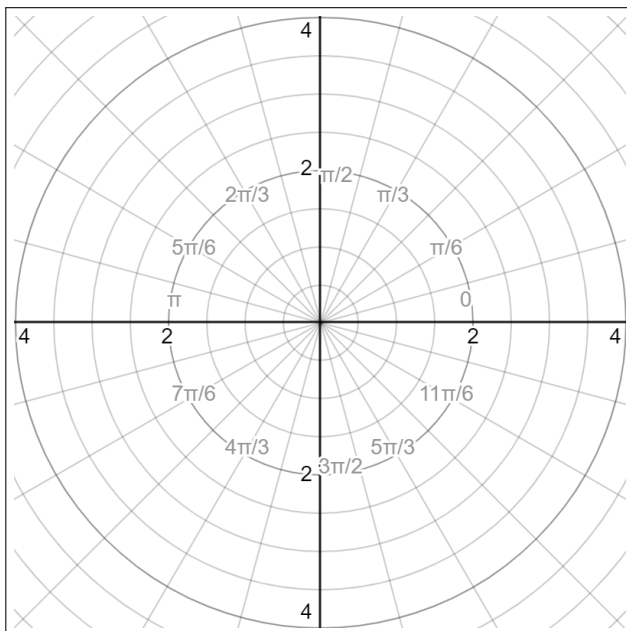


- (b) Consider the lamina in the plane of density $\rho = 1$ bounded by the curves $y = x^3 - x$ and $18x = y^2$. These curves intersect at two points where $y = 0$ and $y = 6$. Find each of the following. You can express your answers using unsimplified and unevaluated integrals.
- (i) The moment of the lamina about the y -axis (ii) The y -coordinate of the center of mass

B5 - Areas and lengths in polar coordinates (inspired by past final worksheet problem 2 and Brightspace problem 10)

Consider the polar curve $r = 1 + 2 \cos \theta$.

- (a) Sketch the polar curve on the grid to the right. Clearly mark any intercepts with the x and y -axes.



- (b) Setup (but don't evaluate) an integral representing each of the following:

- (i) The total length of the polar curve
- (ii) The area bounded by the polar curve in the second quadrant (i.e. $x \leq 0$ and $y \geq 0$)